



Side-stream anaerobic membrane bioreactor system for energy-positive sewage treatment

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ARTICLE INFO

Keywords:

Anaerobic membrane bioreactor (AnMBR)
Chemically enhanced primary treatment (CEPT)
Low-strength wastewater
Bioenergy
Carbon neutrality

ABSTRACT

Anaerobic membrane bioreactors (AnMBRs) present a promising solution for developing energy-neutral wastewater treatment systems. However, their application in mainstream settings, where they treat sewage directly, faces challenges such as dissolved methane loss and performance deterioration at low temperatures. To address these issues, this study developed a side-stream mode AnMBR system by integrating chemically enhanced primary treatment (CEPT) with AnMBR. In this system, sewage is first concentrated in a CEPT unit, and the concentrates are then digested in an AnMBR. A laboratory-scale mesophilic AnMBR, fed with CEPT sludge from a full-scale plant, was operated to demonstrate the design and verify its performance. The results showed that the CEPT-AnMBR system removed approximately 65.7 % of the COD from sewage, converted roughly 51.4 % into methane, with a methane yield of 0.26 L-CH₄/g-COD. By altering the waste flow from low-strength, high-volume sewage to high-strength, low-volume sludge, the side-stream AnMBR system reduced dissolved methane to <1 % of the generated methane, minimized membrane requirement to approximately 0.56 m²/m³-sewage/day, and achieved energy positivity at 0.61 kWh/m³-sewage. These findings demonstrate a successful “capture + digestion” strategy for energy-saving sewage treatment, advancing the application of AnMBR technology in regions with low-temperatures.

1. Introduction

With an increasing emphasis on the developing carbon-neutral sewage treatment, there is a pressing need to develop and implement anaerobic biological treatment process that save operating energy and simultaneously generate bioenergy from sewage (Loosdrecht and Brdjanovic, 2014). In this context, anaerobic membrane bioreactors (AnMBRs) have emerged as a promising solution. As an advanced anaerobic reactor, AnMBR employs membranes to separate liquid from solids, thereby decoupling the hydraulic retention time (HRT) and solids retention time (SRT) (Kim et al., 2011). This capability allows AnMBR to handle low-strength sewage at short HRTs while maintaining sufficient SRT for organic matter (OM) degradation. Consequently, implementing AnMBR to sewage treatment offers benefits including energy savings, bioenergy generation, reduced land footprint, and hence reduced carbon

emissions (Rong et al., 2022b; Lei et al., 2024; Robles et al., 2022; Lim et al., 2019).

In mainstream applications of AnMBR, where AnMBR treats sewage directly, it serves dual functions as both an OM capturer and a bioenergy generator, featuring a compact and simplified structure. However, existing studies, particularly pilot-scale investigations (Kim et al., 2011; Rong et al., 2022b; Chen et al., 2021; Lei et al., 2024; Robles et al., 2022; Lim et al., 2019), suggest that mainstream AnMBRs face several challenges in practical applications. Specifically, sewage is a kind of low-strength, high-volume wastewater, which requires to be treated at minimal HRTs, typically between 6 and 12 h (Shin and Bae, 2018). These characteristics necessitate that mainstream AnMBRs have a high membrane filtration capacity, resulting in a significant membrane-associated capital and maintenance costs (De Vela, 2021). Additionally, the high proportion of particulate OM in sewage and the

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<https://doi.org/10.1016/j.watres.2025.124321>

Received 22 April 2025; Received in revised form 6 July 2025; Accepted 29 July 2025

Available online 30 July 2025

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